Corrected Pinhole Collimator Geometry

Full 3D Derivation for GateParametrisedPinholeCollimator

Corrected from Kochebina (2022) — Equations (9)–(10) replaced with unified 3D treatment

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1 Known Parameters and Coordinate System

Each pinhole is defined by the following parameters read from the `.pin` file and collimator macro settings:

d	diameter of the pinhole at its waist
α	half-angle of the opening cone
$(x,\, y)$	center of the pinhole on the collimator plate
$(x_f,\, y_f)$	focal coordinates on the FOV center plane
z	rotation radius (distance from FOV center to collimator plate)
h	half-thickness of the collimator plate

The pinhole's cone axis runs from the pinhole center $(x,\, y)$ on the collimator surface to the focal point $(x_f,\, y_f)$ at the field-of-view (FOV) center, spanning the rotation radius z. Because the axis is tilted in *both* the x- and y-directions simultaneously, we must use a single, unified 3D tilt angle β in all subsequent derivations.

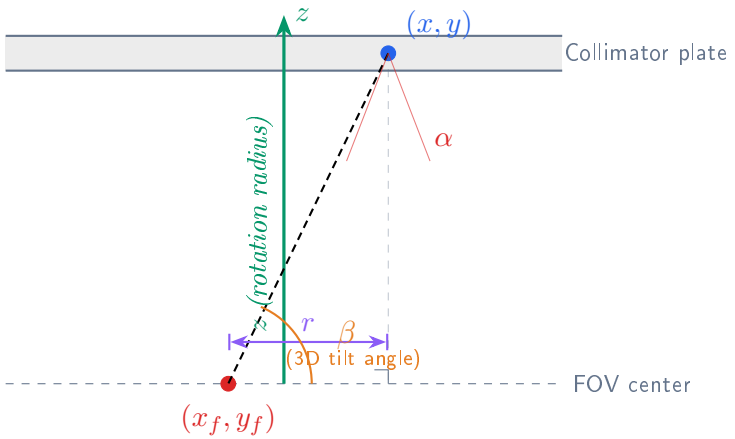


Figure 1: 3D coordinate system. The pinhole at (x, y) on the collimator plate has its axis directed toward the focal point (x_f, y_f) on the FOV center plane, a distance z (rotation radius) below. The 3D tilt angle β is measured from the FOV plane.

2 3D Tilt Angle (Corrected β)

The original document defines (Eq. 1):

$$\beta = \left| \frac{\pi}{2} - \arctan\left(\frac{|x - x_f|}{z} \right) \right|$$

which considers only the x-displacement. In 3D, the total transverse offset from the focal point to the pinhole projection is:

$$r = \sqrt{(x - x_f)^2 + (y - y_f)^2} \tag{1}$$

and the angle β between the cone axis and the FOV-center plane becomes:

$$\boxed{\beta = \arctan\left(\frac{z}{r}\right) = \arctan\left(\frac{z}{\sqrt{(x - x_f)^2 + (y - y_f)^2}} \right)} \tag{2}$$

When $y = y_f$ (no y-displacement), this reduces exactly to the original Eq. (1). The azimuthal angle φ of the tilt direction (needed later for displacement corrections) is:

$$\varphi = \operatorname{atan2}(y - y_f,\ x - x_f) \tag{3}$$

3 Geometrical Cone Parameters

Below we re-derive all cone geometry parameters using the corrected 3D tilt angle β. See fig. 3 for the labeling of geometric quantities. All equations below use the corrected β from eq. (2).

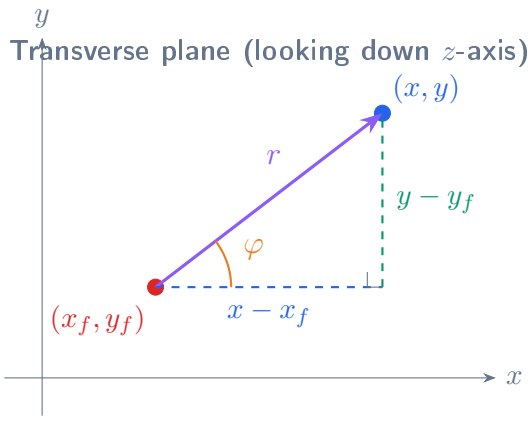


Figure 2: View looking down the z-axis (along the rotation radius). The total transverse offset r combines both x- and y-displacements. The azimuthal angle φ defines the tilt direction.

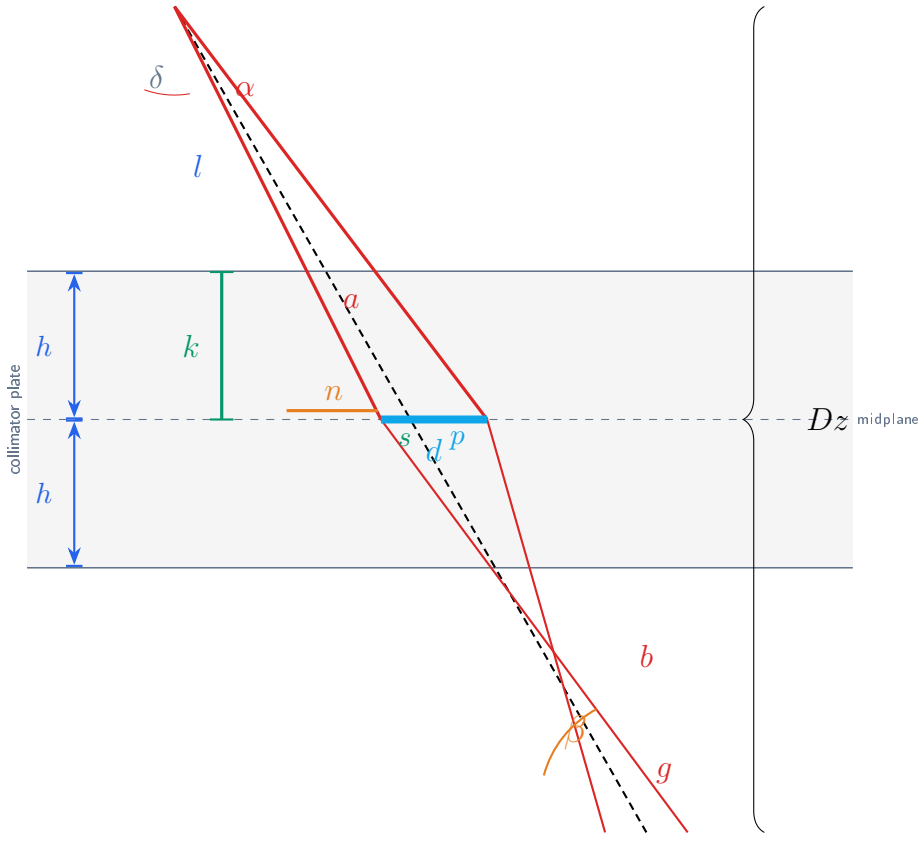


Figure 3: Cross-section of a tilted pinhole cone along the tilt plane. The cone axis makes angle β with the FOV plane. The aperture diameter is d at the midplane, and the collimator half-thickness is h. Labeled quantities: l (upper extension beyond plate), a (midplane-to-plate along axis), k (vertical projection), n (horizontal offset), b (lower extension), g (lower horizontal run). Total cone height: $Dz = a + l + b$.

3.1 Step 1: Auxiliary angle δ

The angle δ between the collimator surface normal (vertical) and the cone axis satisfies $\delta+\alpha+\beta = 90°$, giving:

$$\delta \;=\; 90° - \alpha - \beta \tag{4}$$

3.2 Step 2: Extension above the plate — l

The segment l runs from the upper plate surface along the cone edge to the cone apex. From the right triangle formed by h, the plate surface, and the cone edge:

$$l \;=\; \frac{h}{\cos(\delta+\alpha)} \;=\; \frac{h}{\sin\beta} \tag{5}$$

3.3 Step 3: Vertical and horizontal projections — k and n

Using the two cone edges (at angles δ and $\delta + 2\alpha$ from vertical) and the pinhole diameter d at the midplane, we set up the simultaneous equations:

$$\tan(\alpha+\beta) = \frac{k}{n} \tag{6}$$

$$\tan(\beta-\alpha) = \frac{k}{n+d} \tag{7}$$

Eliminating n by solving eq. (6) for $n = k/\tan(\alpha+\beta)$, substituting into eq. (7), and solving for k:

$$k \;=\; \frac{d\,\tan(\beta-\alpha)\,\tan(\alpha+\beta)}{\tan(\alpha+\beta)-\tan(\beta-\alpha)} \tag{8}$$

$$n \;=\; \frac{k}{\tan(\alpha+\beta)} - d \tag{9}$$

Note

The derivation of k and n follows from the simultaneous equations $\tan(\alpha+\beta) = k/n$ and $\tan(\beta-\alpha) = k/(n+d)$. Subtracting eliminates n and yields the expression for k. Note that n in eq. (9) uses $\tan(\alpha+\beta)$, not $\tan(\beta-\alpha)$.

3.4 Step 4: Axis segment from midplane to upper plate — a

$$a \;=\; \frac{k}{\cos(\delta+\alpha)} \;=\; \frac{k}{\sin\beta} \tag{10}$$

3.5 Step 5: Lower offset quantities — s, p, g, and b

$$s = \frac{k}{\tan\beta} - n \tag{11}$$

Then $p = d - s$. By similar triangles between the upper and lower extensions:

$$g = \frac{(d-s)(a+l)}{a} \tag{12}$$

$$b = g\cos\beta = \frac{(d-s)(a+l)\cos\beta}{a} \tag{13}$$

3.6 Step 6: Total cone height Dz (corrected)

This is the key correction. The original document attempts to split Dz into separate x- and y-components (Eqs. 9–10) and combine them via:

$$\underbrace{Dz_x = a + l + b}_{\text{Eq. (8), using } x\text{-only } \beta}, \qquad \underbrace{Dz_y = \frac{y - y_f}{\cos\big((y-y_f)/z\big)}}_{\text{Eq. (9), dimensionally incorrect}}, \qquad \underbrace{Dz = \sqrt{Dz_x^2 + Dz_y^2}}_{\text{Eq. (10), invalid Pythagorean sum}}$$

Both Eq. (9) and Eq. (10) are incorrect:

- **Eq. (9):** The argument to cos is a dimensionless ratio of lengths, not an angle. The formula also ignores the cone opening angle α and all the careful trigonometry from the cross-section.
- **Eq. (10):** Dz_x and Dz_y are both scalar depths along the cone axis—they are not orthogonal vector components. A Pythagorean combination of two cone heights is geometrically meaningless.

The correct result, using the 3D tilt angle β from eq. (2) throughout Eqs. (5)–(13), is simply:

$$\boxed{Dz = a + l + b} \tag{14}$$

This single equation replaces the original Eqs. (8), (9), and (10).

Expanding fully:

$$Dz = \underbrace{\frac{k}{\sin\beta}}_{a} + \underbrace{\frac{h}{\sin\beta}}_{l} + \underbrace{\frac{(d-s)(a+l)\cos\beta}{a}}_{b} \tag{15}$$

3.7 Step 7: Cone basis radius

$$r_{\max} = Dz \cdot \tan\alpha \tag{16}$$

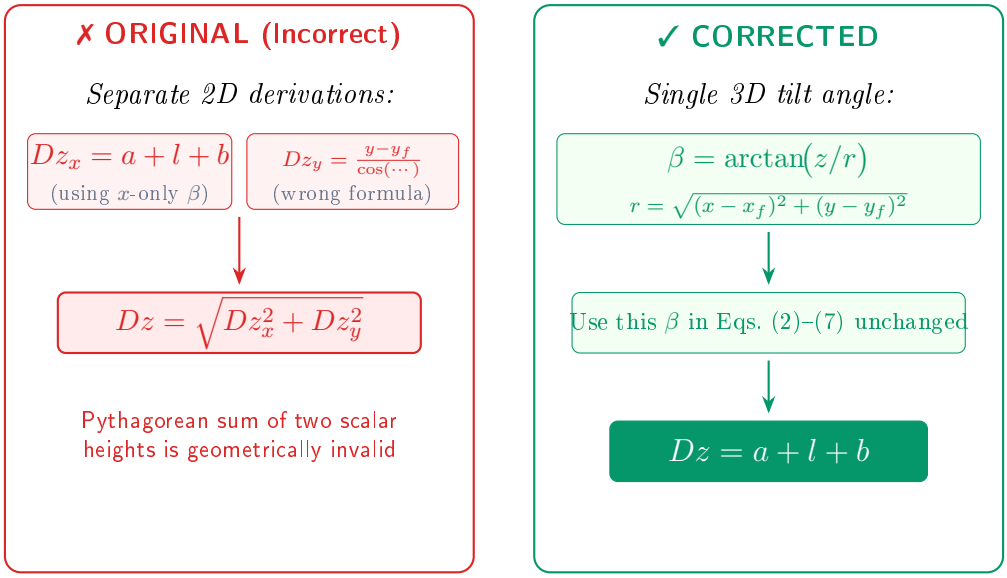


Figure 4: Comparison of the original (incorrect) decomposition vs. the corrected unified approach.

4 Cone Tilt (Rotation Matrix)

The tilted cone is oriented using `G4RotationMatrix` with rotations `rotateX` and `rotateY`. The direction cosines of the cone axis are computed from the displacement vector $(x-x_f,\ y-y_f,\ z)$:

$$t \;=\; (x-x_f)^2 + (y-y_f)^2 + z^2 \tag{17}$$

$$x_{\text{normal}} = \frac{x-x_f}{\sqrt{t}} \tag{18}$$

$$y_{\text{normal}} = \frac{y-y_f}{\sqrt{t}} \tag{19}$$

$$z_{\text{normal}} = \frac{z}{\sqrt{t}} \tag{20}$$

The rotation angles that align the cone axis with this direction:

$$\text{rotX} \;=\; \arctan\left(\frac{x_{\text{normal}}}{z_{\text{normal}}}\right) \tag{21}$$

$$\text{rotY} \;=\; -\arctan\left(\frac{y_{\text{normal}}}{z_{\text{normal}}}\right) \tag{22}$$

Note

These rotation equations are unchanged from the original document. They already operate in 3D since the direction cosines naturally incorporate both x- and y-displacements.

5 Displacement Correction Due to Tilt

Because the `G4Cons` is centered at its geometric midpoint while the pinhole center (x, y) is defined at the collimator midplane, we must compute displacement corrections $(\Delta x, \Delta y, \Delta z)$ to position the cone correctly.

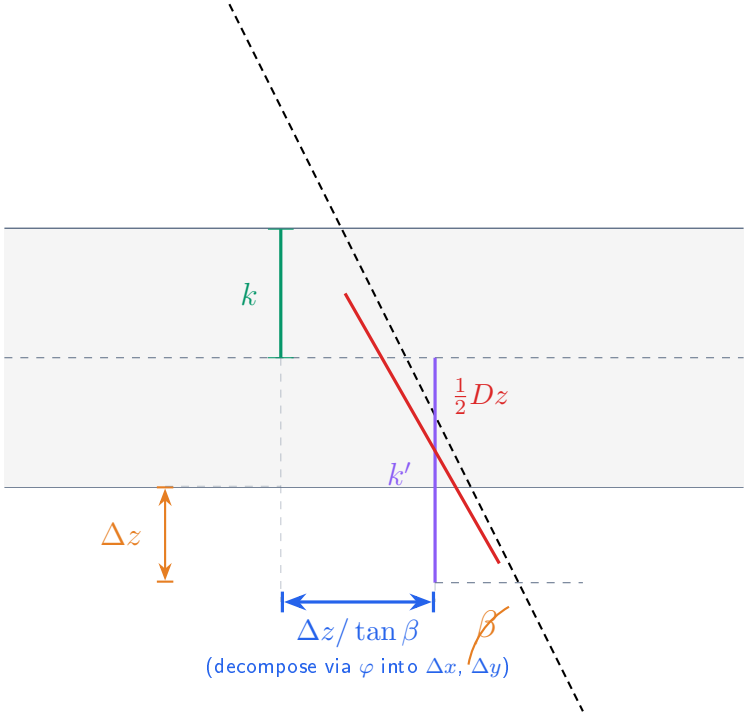


Figure 5: Displacement corrections. The cone center (at $\frac{1}{2}Dz$ along its axis) is offset from the pinhole center at the midplane by $(\Delta x, \Delta y, \Delta z)$.

5.1 Vertical displacement Δz

The cone's geometric center sits at half its total height along the axis. Project this back to the vertical:

$$k' \;=\; \tfrac{1}{2}\,Dz \cdot \sin\beta \tag{23}$$

$$\Delta z \;=\; k - k' \;=\; k - \tfrac{1}{2}\,Dz\,\sin\beta \tag{24}$$

5.2 Transverse displacements Δx and Δy (corrected)

The horizontal offset $\Delta z/\tan\beta$ must be decomposed into its x- and y-components using the azimuthal angle $\varphi = \operatorname{atan2}(y - y_f,\; x - x_f)$:

$$\boxed{\Delta x \;=\; \frac{\Delta z}{\tan\beta}\,\sin\varphi} \tag{25}$$

$$\boxed{\Delta y \;=\; \frac{\Delta z}{\tan\beta}\,\cos\varphi} \tag{26}$$

5.3 Application to upper and lower cones

Each pinhole is drilled from both sides, so there are two cones per pinhole:

Down cone (toward FOV):

$$\begin{cases} x_{\text{down}} = x + \Delta x \\ y_{\text{down}} = y + \Delta y \\ z_{\text{down}} = -\Delta z \end{cases} \tag{27}$$

Upper cone (away from FOV):

$$\begin{cases} x_{\text{up}} = x - \Delta x \\ y_{\text{up}} = y - \Delta y \\ z_{\text{up}} = +\Delta z \end{cases} \tag{28}$$

6 Summary of All Corrected Equations

Table 1 collects every equation in the derivation. Items marked with ✓ are corrected relative to the original document; all others are unchanged in form but now use the corrected β.

Table 1: Complete equation summary. Green rows indicate corrections.

Eq.	Quantity	Formula	Status
green!8 (1′)	β	$\arctan\!\big(z \,/\, \sqrt{(x-x_f)^2+(y-y_f)^2}\big)$	✓ **corrected**
(2)	l	$h/\sin\beta$	unchanged
(3)	k	$\dfrac{d\,\tan(\beta-\alpha)\,\tan(\alpha+\beta)}{\tan(\alpha+\beta)-\tan(\beta-\alpha)}$	unchanged
(4)	n	$k/\tan(\alpha+\beta)-d$	unchanged
(5)	a	$k/\sin\beta$	unchanged
(6)	s	$k/\tan\beta-n$	unchanged
(7)	b	$(d-s)(a+l)\cos\beta\,/\,a$	unchanged
green!8 (8)	Dz	$a+l+b$	✓ **replaces (8)+(9)+(10)**
(11)	$r_{\max}$	$Dz\cdot\tan\alpha$	unchanged
(12–17)	tilt rotations	(see §4)	unchanged
(21)	Δz	$k-\frac{1}{2}Dz\sin\beta$	unchanged
green!8 (22)	Δx	$\sin\varphi\cdot\Delta z/\tan\beta$	✓ **uses φ**
green!8 (23)	Δy	$\cos\varphi\cdot\Delta z/\tan\beta$	✓ **uses φ**

Key insight: The original equations (9) and (10) attempted a 2D+1D decomposition that is mathematically invalid. The fix is conceptually simple: compute the true 3D tilt angle β from the full displacement vector $(x - x_f,\, y - y_f,\, z)$, then use the *same* 2D cross-sectional geometry (Eqs. 2–7) in the tilt plane. The result is a single scalar cone height $Dz = a + l + b$ that correctly accounts for tilt in any direction.